Quantum sampling for a long/short stock split:
a same-budget benchmark of QAOA variants on 12–29 qubits

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Abstract

We encode the choice of a low-risk long/short (or buy/skip) stock portfolio as an Ising problem with one qubit per asset, enumerate the exact optimum at 12 assets, and benchmark seven circuit families (linear-ramp QAOA, free-angle QAOA with five optimisers, CVaR, multi-angle, counterdiabatic, two imaginary-time methods and a constraint-preserving XY mixer) on an exact statevector simulator against classical heuristics *under the same budget*: 1 000 measured shots versus 1 000 evaluated candidate portfolios. Tuned circuits find the optimum in 99–100 % of runs, restarted greedy descent in 82 and 100 % (modes A and B), a 1 000-step annealing run in 47 and 100 %, random sampling in 22 %. The circuits, however, first cost 300–10 700 offline simulated circuits to tune, a price the heuristics do not pay, and at 12 qubits 1 000 greedy evaluations already cover a quarter of the search space. Two findings are robust: the linear-ramp recipe as commonly written has the wrong sign for the $R_x(+2\beta)$ mixer and steers towards the worst portfolios, and angles tuned at 12 qubits transfer to 16 and 20 qubits without loss. We also measure the all-to-all connectivity advantage (66 vs. 285 two-qubit gates per layer at 12 qubits, 244 vs. 1 576 at 29). All optimised portfolios reduced out-of-sample risk; none beat the market's return. No quantum advantage is claimed and no hardware run is reported.

1 Problem and data

Each asset i gets a spin $s_i \in \{+1, -1\}$. Mode A (long/short): $+1$ = buy, -1 = short, weights s_i/n. Mode B (buy/skip): $x_i = (1 + s_i)/2 \in \{0, 1\}$, weight x_i/n, the rest in cash. With training-window mean weekly log returns μ and sample covariance C, the score to minimise is

$$J = \frac{w^{\mathsf{T}} C\, w}{V_{\mathrm{ref}}} - \eta\, \frac{\mu^{\mathsf{T}} w}{\sigma_{\mathrm{ref}}}, \qquad V_{\mathrm{ref}} = \overline{\mathrm{diag}\, C},\ \sigma_{\mathrm{ref}} = \sqrt{V_{\mathrm{ref}}}, \tag{1}$$

with $\eta \in \{0.25, 0.5, 1, 2\}$ ($\eta = 1$ default) and an optional balance constraint (equal numbers on each side). J is an Ising energy $\sum_{i<j} J_{ij} s_i s_j + \sum_i h_i s_i + c$ exactly (verified to 10^{-16} on random states); thinning the 29-asset couplings to the strongest 60 % by $|\rho_{ij}|$ gives a surrogate within 0.013 of the true score (rank correlation 0.975), and every candidate is re-scored on the dense objective.

Data. Kaggle "Huge Stock Market Dataset" (CC0): 29 named US assets in 10 groups (tech, gold & silver, energy, banks, bonds, healthcare, consumer staples, utilities, real estate, copper) plus SPY as a market reference, weekly closes 2006-05-26 to 2017-11-10 (599 weeks, 598 weekly log returns). Chronological split: 418 training weeks (to 2014-05-30, containing the 2008 crash) and 180 held-out weeks; nothing was chosen with test data. The 12-asset instance takes one asset per group in a fixed round-robin: AAPL, GLD, XOM, JPM, TLT, JNJ, PG, XLU, VNQ, FCX, MSFT, SLV. Bonds are the only group negatively correlated with the rest (mean $\rho \approx -0.22$; TLT–XLK -0.45); gold is near zero ($+0.10$ to $+0.14$). An earlier thinning rule by $|C_{ij}|$ was dropped because it removed every bond link.

2 Methods

Answer key and classical baselines. At 12 assets all $2^{12} = 4\,096$ portfolios are enumerated (exactly one optimum per setting). Baselines, written in NumPy: uniform random sampling, greedy single-flip (or swap) descent with restarts, and simulated annealing. Balance is enforced by a penalty $\lambda(\sum_i s_i)^2$ at 12 assets, with λ verified by enumeration so that no unbalanced state beats the best balanced one, and by a hard constraint in the solvers at 29.

Circuit. $H^{\otimes n}$, then per layer k: $R_{zz}(2\gamma_k J_{ij}/\mathrm{scale})$ on every kept link, $R_z(2\gamma_k h_i/\mathrm{scale})$ on every asset, $R_x(2\beta_k)$ on every qubit, with one common scale $= \max(|J_{ij}|, |h_i|)$; the constant is a global phase. Linear-ramp QAOA (LR-QAOA) [1] uses $\gamma_k = \frac{k}{p}\Delta\gamma$, $\beta_k = (1 - \frac{k-1}{p})\Delta\beta$ and was grid-tuned (300 points per p, $p = 1..5$) by lowest expected energy, never by the optimum's probability, because the optimum is unknown in a real problem. Free-angle QAOA ($2p$ angles) was tuned by COBYLA, Nelder–Mead, Powell, L-BFGS-B and our own SPSA from the same five starts (the default ramp, the tuned ramp, three fixed-seed random points) with a cap of $200p$ circuit evaluations each. Extras at the same cap: a CVaR objective ($\alpha = 0.1, 0.25$), multi-angle QAOA ($3p$: separate angles for links and fields), a first-order counterdiabatic R_y layer (DC-QAOA, cf. [3]), VarQITE angle training (McLachlan, central differences), varQITE with the Pauli-term update rule of IonQ/ORNL [2] (ansatz A: R_y + CX per link, 66 parameters; ansatz B: R_{zz} per link + R_x layer; implemented from the method description, no code reused), and an XY mixer with a Dicke initial state for the balance constraint. An independent NumPy statevector simulator (1.7 ms per $p = 2$ circuit at 12 qubits) agrees with Qiskit 2.5.2 `Statevector` to 3×10^{-16} for all variants; the R_{zz}/R_z sign convention is unit-tested against the explicit diagonal unitary. 122 unit tests pass; the full benchmark ran in 121 s on 8 cores.

Same-budget protocol. One *candidate evaluation*

is the energy of one portfolio: a measured shot, a random sample, one neighbour examined by greedy descent, or one annealing proposal. Every method gets exactly $B = 1\,000$ of them; hit@1000 is the probability that the optimum is found within the budget (exact $1-(1-P_{\text{opt}})^{1000}$ for circuits and confirmed by 200 sampled repetitions; 200 repetitions for the heuristics). Greedy restarts from random states until the budget is used (19 and 12 restarts on average in modes A and B); annealing runs one geometric schedule of 1 000 proposals. The circuits' offline tuning cost is listed separately.

3 Same-budget benchmark

Table 1 and Fig. 1 give the fair comparison. With 1 000 evaluations each, the tuned circuits reach the optimum in 99–100 % of runs in both modes; restarted greedy reaches 82 % (A) and 100 % (B), the 1 000-step annealer 47 % and 100 %, uniform sampling 22 %. Three caveats keep this honest: (i) the circuits were first tuned with 300 (LR-QAOA) to 10 685 (free-angle, all optimisers and starts) simulated circuits, which the heuristics never needed; (ii) at 12 qubits 1 000 greedy evaluations already cover a quarter of the 4 096 states, and the annealer's schedule was not tuned for such a short run; (iii) all distributions are exact and noiseless. Mode B is easy for every method because its optimum ("buy AAPL and TLT", everything else in cash) is reached by few flips from most states; mode A, a genuine hedge, separates the methods.

4 Per-shot results and the two findings

Per shot, the tuned circuits are 30× (A) and 85× (B) more likely than uniform sampling to return the optimum (Table 1); the CVaR objective reaches 236× in mode B at a lower mean quality. In mode A the circuit's single most likely portfolio *is* the optimum (rank 1 of 4 096: long AAPL, MSFT, GLD, SLV, JPM, TLT, PG, XLU; short XOM, JNJ, VNQ, FCX; Fig. 2); in mode B it is "buy TLT only", one asset from the optimum (rank 6).

Finding 1: the sign of the ramp. With the mixer $R_x(+2\beta)$ and $H^{\otimes n}$ as the initial state, lowering the energy requires γ and β of *opposite* sign. The recipe as commonly written ($\Delta\gamma = 0.6$, $\Delta\beta = 0.3$) drives the state towards the *highest* energies: quality 0.45 at $p = 5$ against 0.81 for uniform sampling in mode A, 0.33 against 0.71 in mode B, and the probability of the worst portfolio grows with p. Fig. 3 shows the grid: the good region lies entirely in the half-plane $\Delta\beta < 0$. All tuned ramps have $\Delta\beta < 0$: A (0.3,−0.3), (0.5,−0.3), (0.7,−0.4), (0.7,−0.4), (0.7,−0.4) for $p =$ 1..5; B (1.0,−0.7), (1.4,−0.6), (1.4,−0.6), (1.4,−0.5), (1.3,−0.5).

Finding 2: angle transfer. The ramp tuned at 12 assets keeps its quality at 16 and 20 assets (Table 2), within 0.005 of a fresh grid search at 16, and still beats uniform sampling at 20 assets thinned to 60 % of the links as the 29-asset model is. This justifies running the 29-qubit circuits with the 12-asset ramp, untuned; beyond about 20 qubits offline tuning is impossible and would have to be paid in hardware shots.

Optimisers and extras. The optimiser matters less than the start: from the tuned ramp all five reach about the same quality (mean over starts and p: A 0.857–0.895, B 0.950–0.969), random starts often stall, and L-BFGS-B needs the fewest evaluations (24–98). CVaR raises P_{opt} at the cost of mean quality; multi-angle gains little; the counterdiabatic layer in mode B concentrates on near-optimal states (quality 0.985) but never on the optimum. VarQITE angle training matches standard QAOA from the same starts. varQITE with the Pauli-term update (Fig. 4) lowers the energy steadily but slowly at the published learning rates (0.05–0.4) and reaches quality 0.880 (A) / 0.871 (B) after 60 steps and 7 980 simulated circuits per run with 66 two-qubit gates, below LR-QAOA at one layer; its IonQ-native ansatz B is a stationary point at $\theta = 0$ (every parameter-shift derivative vanishes) and needs a random start. Exact imaginary-time evolution of a diagonal cost is classical Boltzmann sampling; as a reference curve it overtakes the best LR-QAOA at $\tau = 80$ (A) / 160 (B).

5 Balance constraint in the circuit

The penalty coupling 2λ is 12× (A) / 5× (B) the largest objective term, so penalised circuits mostly learn the constraint and their grids run into the small-$\Delta\gamma$ edge. The XY mixer (12 XX+YY gates per layer on a ring, Dicke state prepared generically, not gate-optimised) keeps every shot balanced and is 10× more likely than the penalty to return the best balanced portfolio (Table 3). At 29 qubits post-selection of the unconstrained circuit is the practical choice.

6 Gate counts

On all-to-all connectivity one layer costs one native two-qubit gate per link; routed on a heavy-hex grid the same layer costs 4.3× (12 qubits) and 6.5× (29 qubits) more (Table 4). The 29-qubit circuits (732 two-qubit gates at $p = 3$) were built and counted only, never simulated.

7 Out-of-sample finance

Over the 180 held-out weeks (simple returns, equal weights, no costs), the classically optimal portfolios of all 32 settings (2 modes × 4 η × balance on/off) had lower weekly volatility than SPY (0.00–0.91 % vs. 1.57 %) and than 1 000 random portfolios of the same kind; 14 of 32 earned more than a random portfolio; none beat SPY's return (+0.165 vs. +0.214 % per week in a strongly rising market). The method reduces risk; it does not raise return. Trading costs, borrow fees and slippage are not modelled, and the universe has survivorship bias.

Table 1: Same-budget benchmark at 12 assets, $\eta = 1$, balance off. hit@1000: probability of finding the optimal portfolio within 1 000 candidate evaluations. P_{opt}: probability that a single shot is the optimum (uniform: $1/4096 = 0.00024$). Quality $= (E_{\max} - \langle E \rangle)/(E_{\max} - E_{\min})$ of the sampled distribution (1 = always the optimum). Tuning: offline simulated circuits used to choose the angles (all optimisers and starts at that p). Best p is given as A / B.

		Mode A: long/short			Mode B: buy or skip		
Method (best p A / B)	tuning	hit@1000	P_{opt}/shot	quality	hit@1000	P_{opt}/shot	quality
uniform random sampling	0	0.217	0.0002	0.807	0.217	0.0002	0.714
greedy descent, restarted	0	0.815	–	–	1.000	–	–
simulated annealing, one 1000-step run	0	0.465	–	–	1.000	–	–
LR-QAOA (5 / 4)	300 / 300	0.999	0.0073	0.950	1.000	0.0207	0.973
free-angle QAOA (3 / 2)	10 685 / 6 489	0.986	0.0042	0.935	1.000	0.0122	0.976
CVaR-QAOA (α=0.1) (3 / 3)	3 047 / 4 860	0.999	0.0066	0.918	1.000	0.0577	0.948
multi-angle QAOA (3 / 2)	5 730 / 3 626	0.995	0.0054	0.945	1.000	0.0141	0.986
counterdiabatic QAOA (3 / 3)	5 154 / 5 210	0.999	0.0067	0.938	$<10^{-3}$	$<10^{-4}$	0.985
VarQITE angle training (3 / 3)	3 000 / 3 000	0.986	0.0042	0.935	1.000	0.0258	0.978
varQITE Pauli-update [2]	38 792 / 46 891	$<10^{-3}$	$<10^{-4}$	0.880	$<10^{-3}$	$<10^{-4}$	0.871

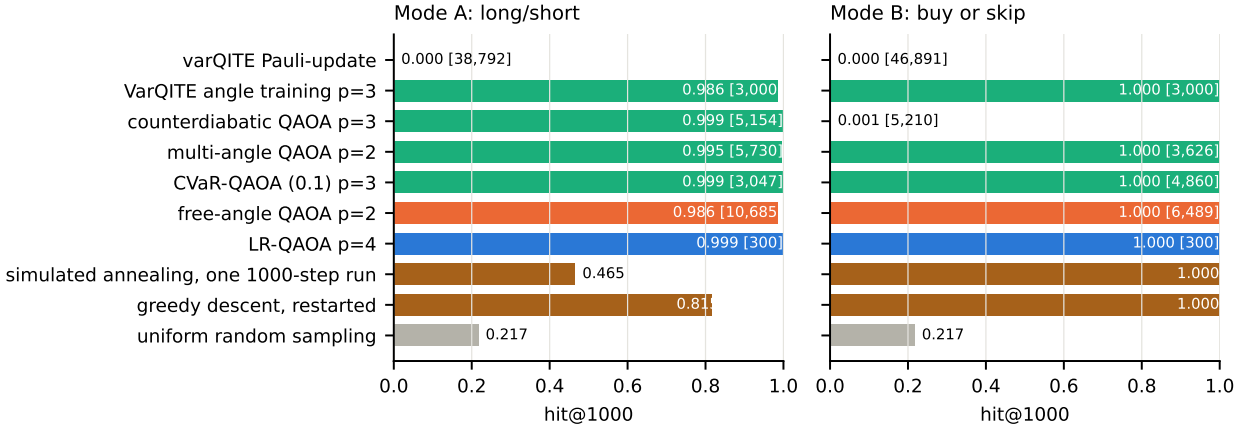


Figure 1: hit@1000 for every method under the same budget of 1 000 candidate evaluations. Brown: classical heuristics (measured over 200 repetitions); grey: uniform sampling; blue, orange, green: circuits (exact). The bracketed numbers are the offline tuning circuits.

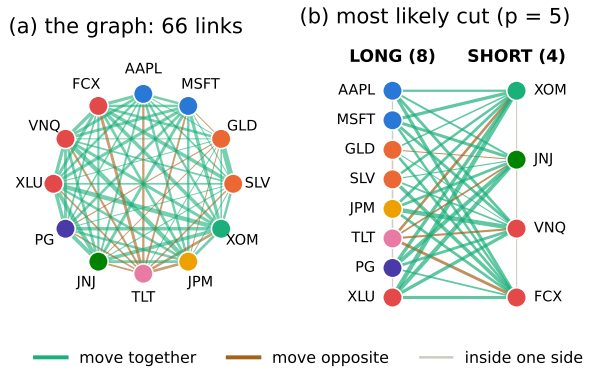


Figure 2: Mode A, 12 assets. (a) Training correlations: teal = move together, brown = move opposite, width $\propto |\rho|$. (b) The most likely LR-QAOA portfolio: 27 of 52 "together" links are cut (hedged) and only 5 of 14 "opposite" links.

Table 2: Transfer of the 12-asset ramp: quality of the sampled distribution at $p = 1/2/3$ with the 12-asset angles ("own": a fresh grid search at that size).

assets (links)	mode	$p = 1$	$p = 2$	$p = 3$
12 (all)	A	0.895	0.916	0.931
16 (all)	A	0.889	0.905	0.922
16 (all), own	A	0.890	0.910	0.926
20 (all)	A	0.906	0.919	0.937
20 (60 %)	A	0.864	0.882	0.900
12 (all)	B	0.955	0.969	0.972
16 (all)	B	0.949	0.960	0.963
16 (all), own	B	0.949	0.960	0.963
20 (all)	B	0.959	0.968	0.969
20 (60 %)	B	0.950	0.963	0.966

8 Limitations and outlook

Everything above is an exact classical simulation; the IonQ noise-model run (one batched job with the $p = 0$ control, LR-QAOA, CVaR and XY circuits; 12 qubits live, 29 recorded) is the next step and will show how much survives 66–244 two-qubit gates per layer. The same-budget comparison favours the circuits only after their offline tuning, which is impossible beyond about 20 qubits; the angle transfer is the way around it and is itself measured only up to 20. Deferred: warm starts, variational real-time evolution, recursive QAOA. Code, 122 tests, the saved angles and all run records (including every unhelpful variant) are in the team repository.

References

[1] J. A. Montañez-Barrera and K. Michielsen, Toward a linear-ramp QAOA protocol: evidence of a scaling advantage in solv-

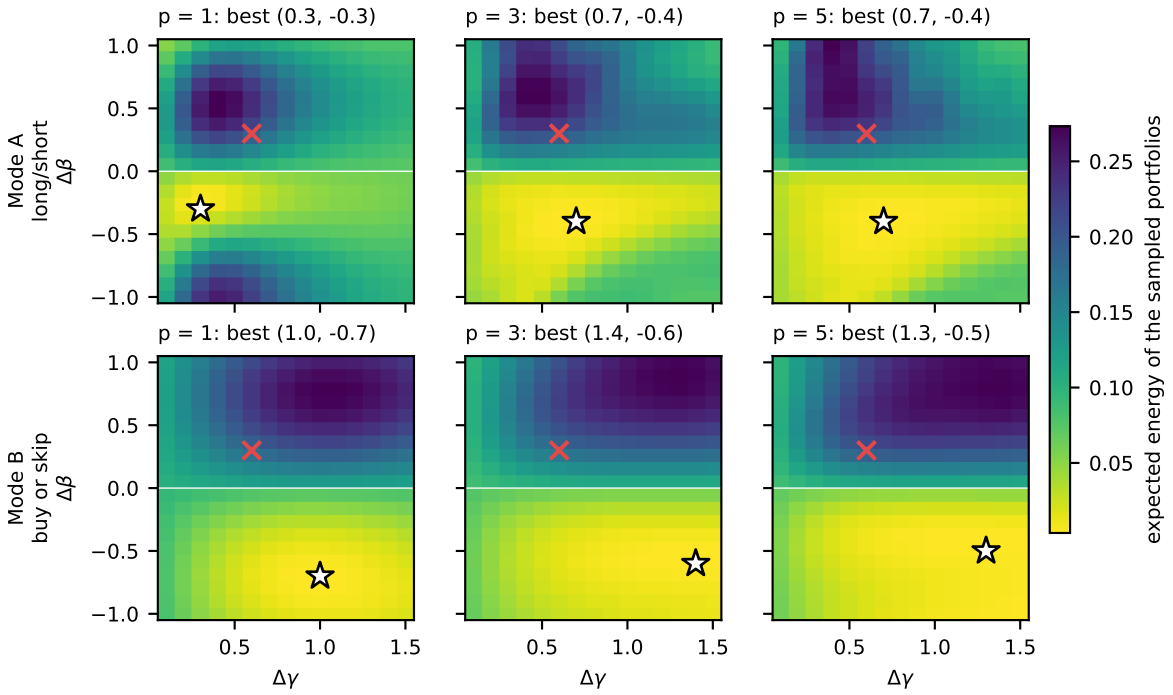


Figure 3: LR-QAOA grid search, expected energy of the sampled portfolios (darker = lower = better) over $\Delta\gamma$ and $\Delta\beta$ for $p = 1, 3, 5$. White star: chosen ramp; red cross: the recipe as commonly written (0.6, 0.3), which lies in the bright upper half where the circuit samples worse-than-random portfolios.

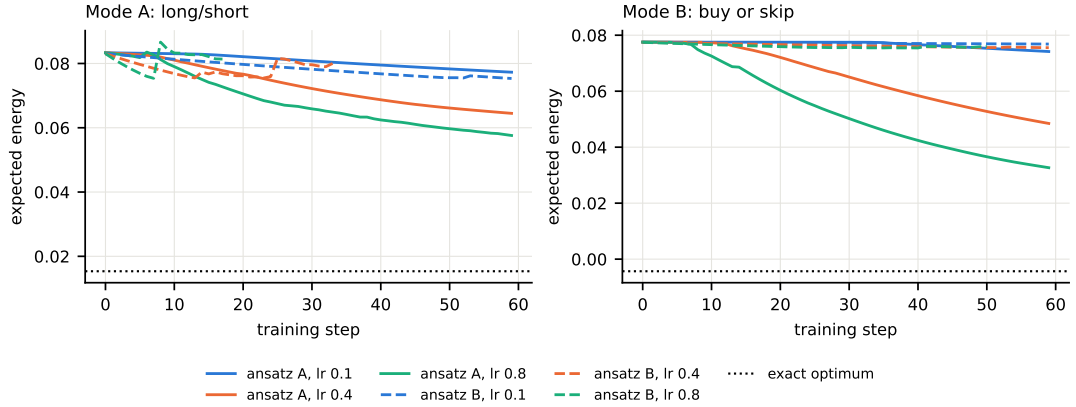


Figure 4: varQITE (Pauli-term update rule [2]) training curves on the exact statevector: expected energy per step for both ansätze (solid: A, dashed: B) and three learning rates (blue 0.1, orange 0.4, green 0.8); dotted: the exact optimum. Ansatz A learns, ansatz B barely moves from its small random start.

Table 3: Equal buys and sells at 12 assets, $p = 2$: share of balanced shots and probability per shot of the best balanced portfolio.

	Mode A		Mode B	
method	balanced	P_{best}	balanced	P_{best}
uniform, post-selected	23 %	0.0002	23 %	0.0002
penalty in the model	97 %	0.0011	97 %	0.0011
post-selection	31 %	0.0019	1 %	0.0009
XY mixer + Dicke start	100 %	**0.0192**	100 %	**0.0068**

Table 4: One QAOA layer, measured with the Qiskit transpiler (optimisation level 1, five routing seeds). All-to-all: trapped ions, where R_{zz} is native on IonQ Forte. Heavy-hex: a superconducting layout with SWAP routing (median, min–max).

qubits	links	2q gates	2q depth	heavy-hex 2q gates
12	66	66	15	285 (276–288)
29	244	244	29	1 576 (1 510–1 597)

ing some combinatorial optimization problems, *npj Quantum Information* **11**, 131 (2025), doi:10.1038/s41534-025-01082-1.

[2] IonQ and ORNL, variational quantum imaginary-time evolution with a Pauli-term update rule, arXiv:2404.16135.

[3] IonQ and Kipu Quantum, portfolio optimisation on IonQ hardware, arXiv:2602.23976.